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Artificial intelligence applications in aviation accident classification: A preliminary exploratory study

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ABSTRACT

Artificial intelligence (AI) applications are already integrated and expected to expand within the aviation sociotechnical ecosystem. Machine Learning (ML) models are capable and could excel on accident report classification tasks within the aviation context, assuming adequate and proper training. Human subject matter experts (SMEs) and the open access AI ChatGPT reanalyzed the TransAsia Airways flight GE235, Air France AF447, and Helios Airways Flight HCY552 accidents using the Human Factors Analysis and Classification System (HFACS) to investigate whether the AI agent is capable of conducting aviation accident analysis and comparing raters' results, of verifying AI's learning abilities, and, finally, investigating whether AI analysis could expand in other systems-based models. The results suggested that: (a) AI/ML applications could be an efficient tool for learning organizations' continuous learning, growth, and improvement by creating new knowledge from previous failures, (b) the proper prompts and well-established frameworks could optimize the agreement between human SMEs and AI in accident analysis, (c) the ChatGPT performed better on linear models (i.e., Bowtie) compared to complex systemic accident causation models (e.g., HFACS), (d) the ChatGPT could not conduct high- and low-level analysis simultaneously, and, (e) a multidisciplinary team including human factors, IT, and accident investigators experts is essential in developing and overseeing the W-shaped learning assurance process.

1. Introduction

Most accidents or incidents are not caused by a single factor but rather by a combination of interacting contributing factors [1]. Human Factors Analysis and Classification System (HFACS) is a systems-based accident framework utilized in the analysis and understanding of the human factors that contribute to accidents or incidents in complex systems, such as aviation, healthcare, and the military [2–4]. HFACS provides a way to systematically identify and categorize these contributing factors so that they can be analyzed and addressed [1,5]. Accident and incident analysis is promoted not only by regulations for threat identification [6–8] but also strengthens complex systems' continuous learning, growth, and improvement as a learning organization [9–11].

A learning organization can improve its ability to adapt to changing circumstances, innovate, and achieve its goals [10,12]. Organizations can improve their safety performance and reduce the risk of accidents or incidents by using appropriate methods and tools to create new knowledge from previous successes and failures [12,13]. Artificial intelligence (AI) applications are already integrated and expected to expand within the aviation sociotechnical ecosystem [14–16]. Machine learning models are capable and, assuming adequate and proper training, could excel on report classification tasks in Level 1 applications [17,18] within the aviation context [14].

Literature suggests that AI applications might have the potentials to contribute on organizational learning through adverse events analysis [19–21], and classification in a timely, efficient [22] and standardized fashion by increasing the validity while decreasing SME dependance [21,23]. This study reanalyzed the TransAsia Airways flight GE235, Air France AF447, and Helios Airways Flight HCY552 accidents using the human HFACS model by two different categories of raters: human subject matter experts (SMEs) and the open access AI ChatGPT [24]. The aim was to investigate whether the AI agent is capable of aviation accident analysis, to compare raters' results, to verify AI's learning abilities, and, finally, to investigate whether AI analysis could expand in other systems-based models.

2. Methodology

The authors acted as human factors analysts in the analysis and classification process for the TransAsia Airways flight GE235 [25], Air France AF447 [26], and Helios Airways Flight HCY552 accidents [27].

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Fig. 1. Analysis method comparison based on systemic, theoretical, methodological, and practical criteria as adapted from Stanton et al. [23] with highlighted the method used for this study.
Notes: EAST–BL: Event Analysis of Systemic Teamwork – Broken Links; FTA: Fault Tree Analysis; FRAM: Functional Resonance Analysis Method; HFACS: Human Factors Analysis and Classification Scheme; STAMP–CAST: Systems–Theoretic Accident Model and Processes – Causal Analysis using Systems Theory; STEP: Sequential Timed Event Plotting.

Criteria for the study cases' accidents were: (a) controlled flight into terrain (CFIT) as causal factor, (b) final reports were released, (c) final reports were published in English language, (d) no previous published HFACS analysis was conducted and been available online on relevant academic databases (based on research and to authors' best knowledge to date of writing).

Stanton et al. [23] compared eight analysis methods based on systemic, theoretical, methodological, and practical criteria (Fig. 1). Specifically, the HFACS in this analysis made themes in causal factors easy to identify, enhanced reliability and could facilitate the further statistical process [28]. Moreover, it was expected that researchers' extensive experience and familiarity with the method would benefit reliability. However, HFACS constrained the classification to domain-specific types of failures [23] and predefined organizational levels [29], ruling out factors outside the involved organization and interactions between contributing factors. Thus, complimentary methods were utilized [5,30,31].

Based on the study's scope, the Person-Level and Mishap-Level Factors from HFACS version 7.0 were used, as adopted by the US Air Force (Air Force Safety Center, 2005) due to in-built classification scheme, ease of use, low-training demand (Fig. 1), and the researchers experience. In detail, Level-1 "Acts" included the "performance-based" or "judgement & decision-making errors" and "violations". Level-2 "Preconditions" included the subcategories "environment" (further divided into "physical" and "technological"), "physical and mental state" (further divided into "physical problems", "state of mind", "sensory misperception", and "mental awareness") and, lastly, "teamwork". Level-3 "Supervision" included the "Supervisory violations", "planned inappropriate operations", and "Inadequate supervision". Level-4, "Organizational influences", included the subcategories "resource problems", "personnel selection & staffing", "policy & process issues", and, lastly, "climate/culture influences".

Both analysts have been accident/incident investigators for the European Military Organization (EMO), each with ten (10) years of experience. They were trained in HFACS according to the EMO's regulations and the syllabus for the respective safety officers' courses. Analysts used the same data sources (i.e., official final reports) to support their unique form of analysis. The accidents were initially coded independently by each analyst following the framework's guidance [32]. Upon completing the initial analyses, the two analysts held Microsoft Teams' online meetings to exchange and review their outputs. Any discrepancies or disagreements were resolved through discussion until a consensus was reached.

Then, the research team prompted sequentially in iterative fashion the ChatGPT version February 2023 [24] to conduct an HFACS analysis for the respective case studies aviation accidents. For cohesion and to facilitate readers' comprehension the detailed prompts are included on the relevant results paragraphs (Sections 3.1.2, 3.2.2, and 3.3.2).

Moreover, ChatGPT prompted to conduct a Bowtie [33], an Accident Modeling- AcciMap [34], and a System- Theoretic Accident Modeling and Processes (STAMP) analysis [35] for TransAsia [13]. *Bowtie analysis* [33] is a risk management methodology that helps visualize an event's cause and effect. It uses a diagram that resembles a bow tie, with the event at the center and the causes and consequences on either side. AcciMap analysis [34] visually represents the events and factors leading up to an accident. STAMP [35] is a structured process for identifying and analyzing aviation safety risks. It consists of four steps: Identify, Analyze, Mitigate, and Evaluate. Finally, AI was prompted to suggest an accident analysis framework and mitigation measures to avoid similar accidents in the future. Similarly, for cohesion and to facilitate readers' comprehension the detailed prompts are included on the relevant paragraphs in Section 4.

3. Case studies results

3.1. TransAsia Airways (TNA) GE235 HFACS analysis

On February 4th 2015, the domestic flight TransAsia Airways (TNA) Flight 235 from Taipei to Kinmen, Taiwan, operated by an ATR 72–600,